Object-Oriented Design
- Design POST

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Design a Project

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- Represent the design of an operation by object sequence diagram
- The object sequence diagram represents the body of the method for the use case operation

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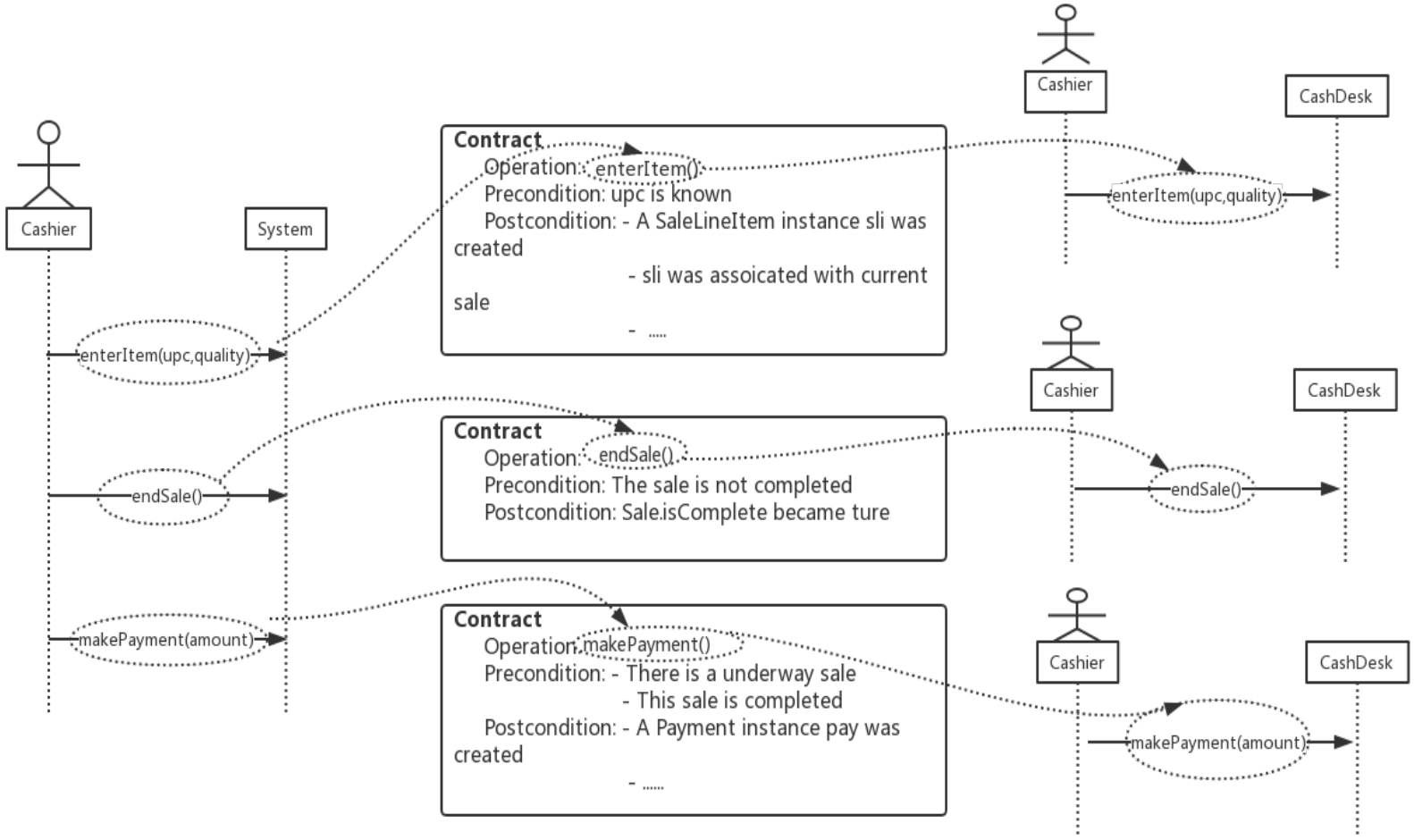
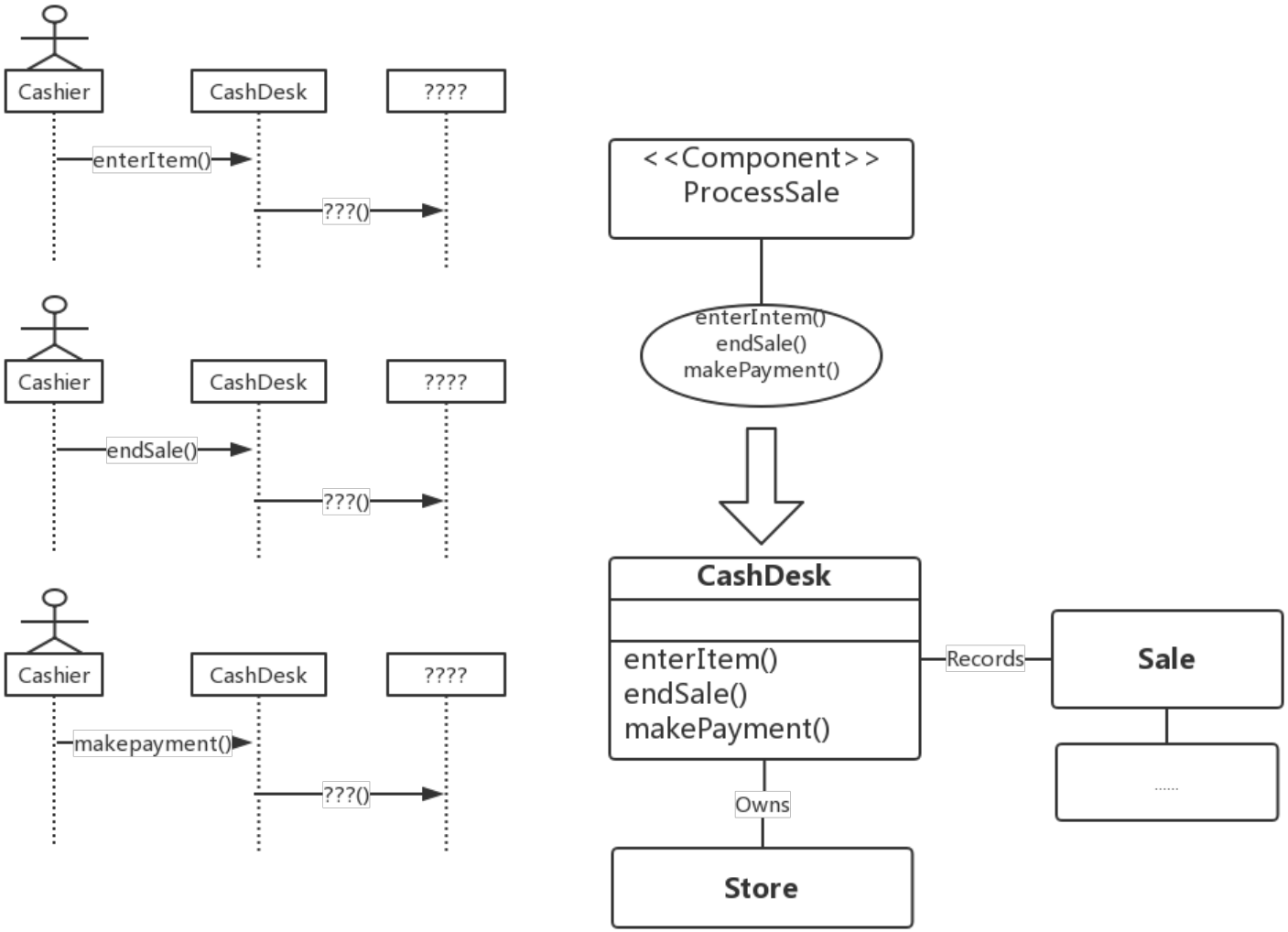
- the contract of the operation, conceptual class diagram, and follow GRASP patterns
- A responsibility assigned to a class is also a method of the class
- The conceptual class diagram is thus refined to a design class diagram
- For each use case, we use an **interface class** as the controller, following the Controller Pattern

Use Case Design Overview



Thanks to Yukun Zhang for the diagram

Overview of Use Case Operation Design



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Design *enterItem*()

Recall its contract

Responsibilities: Record sale of an item, display item description and price.

Pre-conditions: The upc is known to the system.

Post-conditions:

- if a new sale, a *Sale* was created.
- if a new sale, the new *Sale* was associated with the *POST*.
- a *SalesLineItem* was created.
- *SalesLineItem.quantity* was set to *quantity*.
- *SalesLineItem* was associated with a *ProductSpecification*, based on *UPC* match.
- *SalesLineItem* is added to *Sale*

Displaying Item Description and Price

- Model objects are not responsible for display of information
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  - access to data known by model objects, and
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- These are the responsibilities of GUI objects that
  - access to data known by model objects, and
  - can send messages to model objects.
- However, model objects do not send messages to interface objects in order to keep the system independent of the interface.
- This is a matter of interface design and we leave it for later discussion.

- Decide the controller object for *enterItem*(), apply Controller Pattern

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Design methods for postconditions

The post conditions in the contract of *enterItem* include:

- If a new sale, a *Sale* was created.
- If a new sale, the new *Sale* was associated with the *CashDesk*.

Follow Creator Pattern

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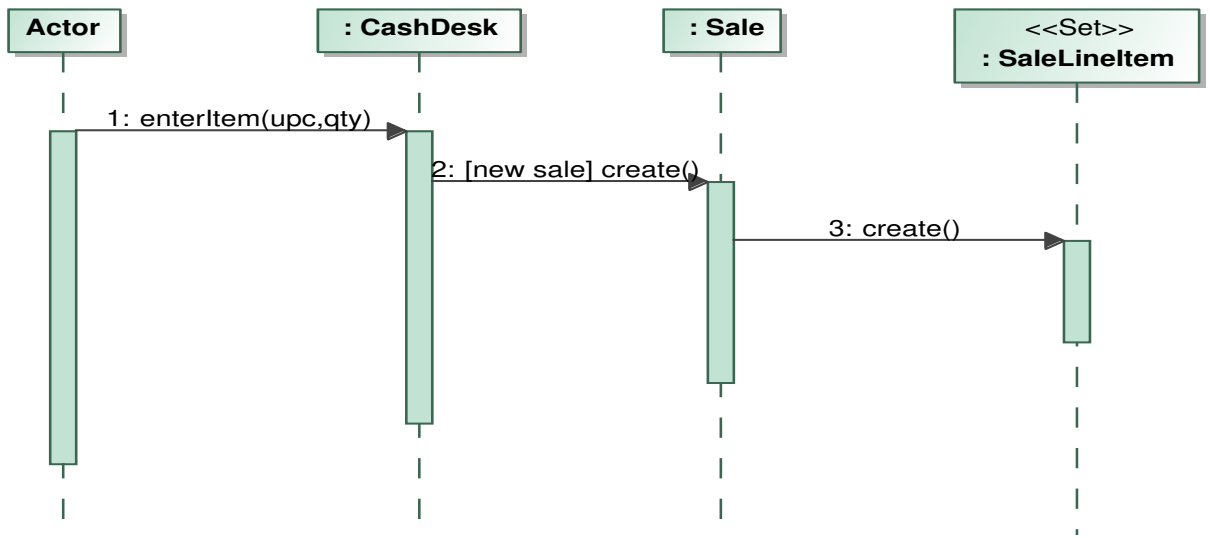

But, is the new Sale associated with *CashDesk*?

Creating a New *SalesLineItem*

Other post-conditions in the contract for *enterItem*() include:

- A *SalesLineItem* was created.
- The *SalesLineItem* was associated with the *Sale*.
- *SalesLineItem.quantity* was set to *qty*.
- The *SalesLineItem* was associated with the *ProductSpecification*, based on upc match.

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Who should be creator?

By Creator Pattern, the answer is *Sale*

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- SalesLineItem has attributes “quantity”, and ”specification” (at least name of item)
- This implies that the creation of *SalesLineItem* needs input values for these attributes
- Thus the message *makeLineItem*() from *CashDesk* to the *Sale* should include ”quantity” and *ProductSpecification*, and
- Sale then passes them in the creation message to the *SalesLineItem*.

Finding a *ProductSpecification*

- where to get *ProductSpectification before* *makeItem*(*spec*, *qty*) is sent to the *Sale*?

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- Notice that the postconditions of *enterItem* also requires that

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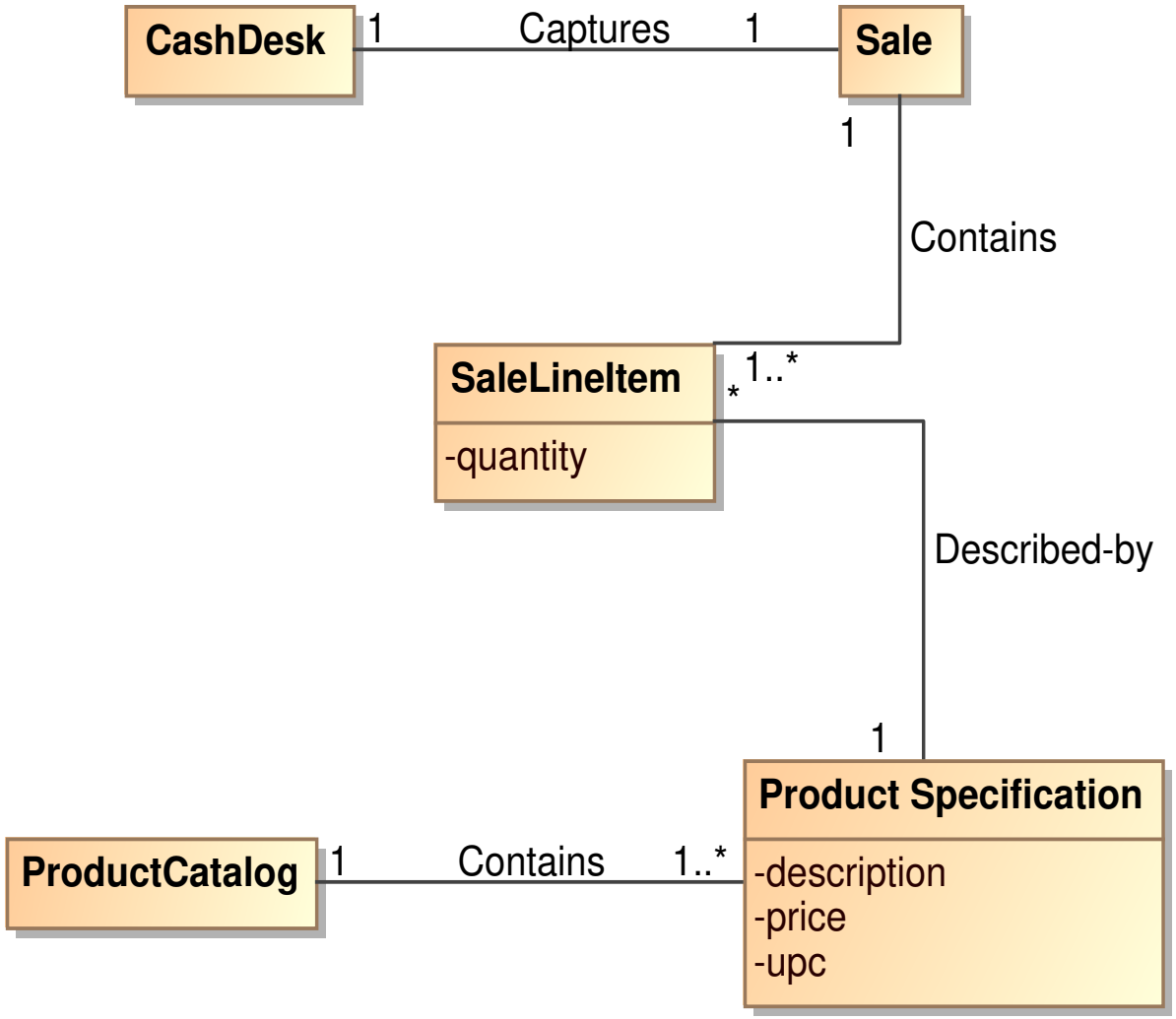
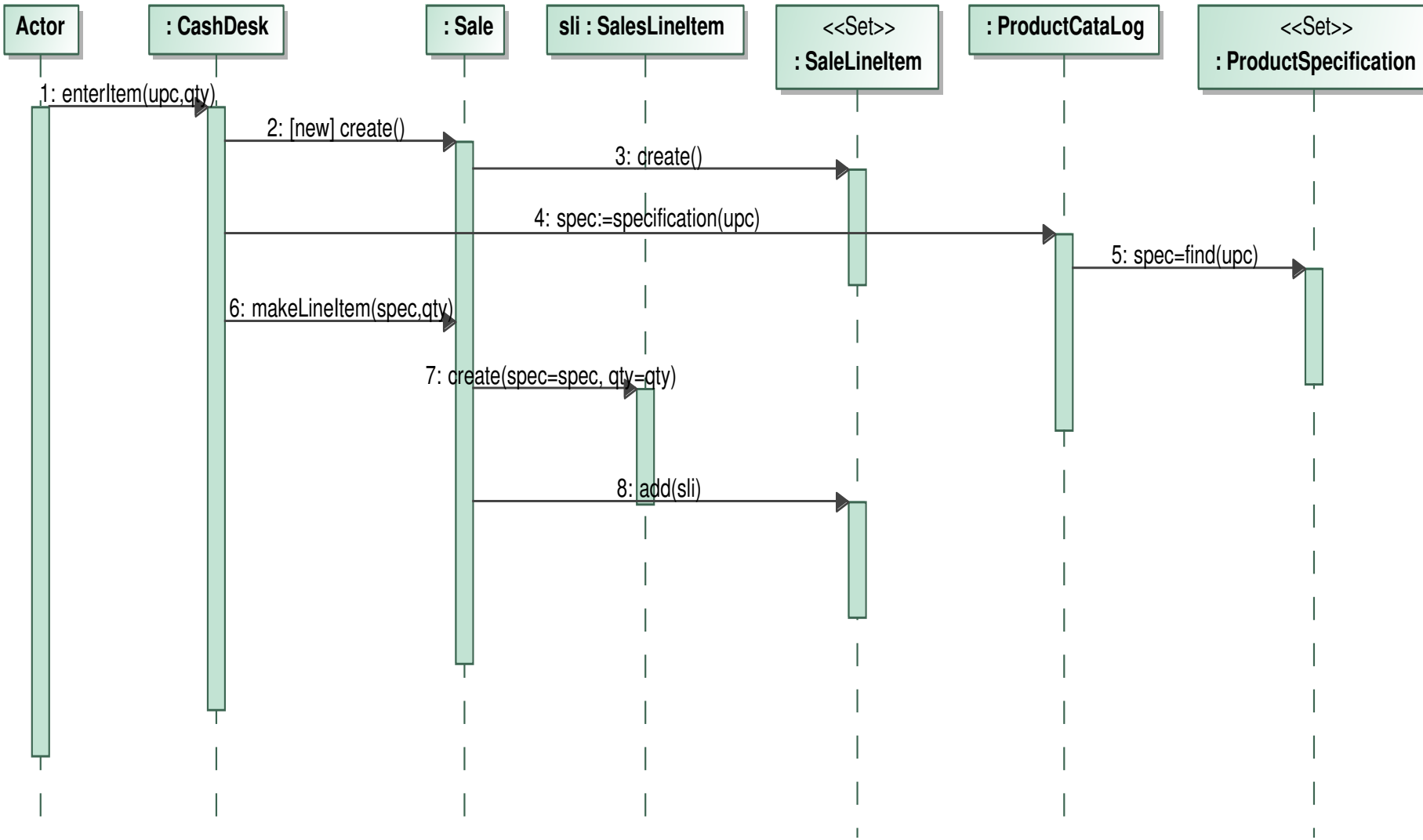
- Thus, the Answer is that *ProductCatalog* should be the expert knowing and looking up a *ProductSpecification*, based on UPC match.

Finally after the SalesLineItem is created, it is associated with the sale, **but how?**

Part of the Conceptual Model

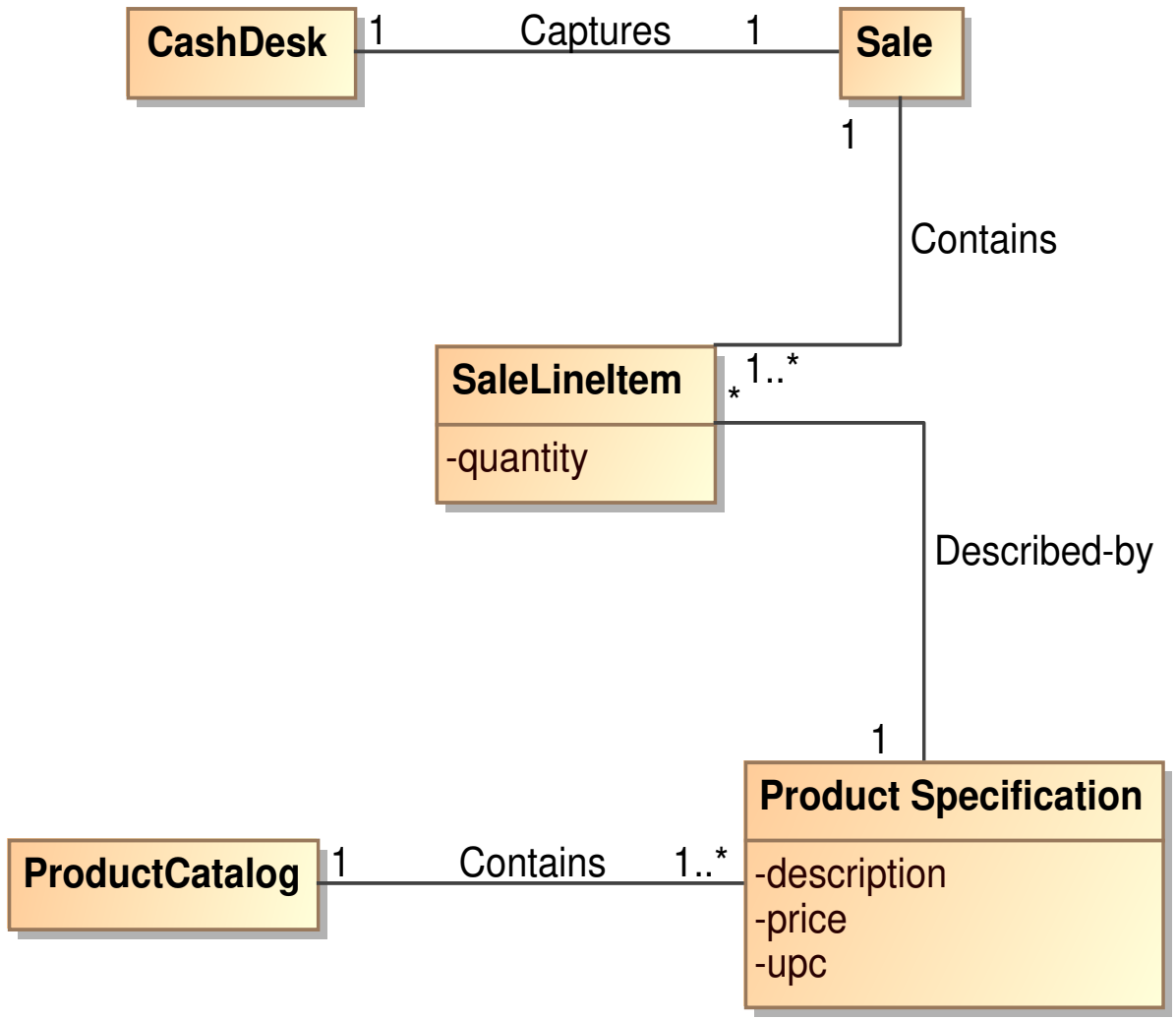


Complete Design of *enterItem(upc, qty)*



Check the Class Diagram Again

Anything missing from the class diagram?



Pseudo-code for *enterItem*(*upc*, *qty*)

CashDesk:: enterItem(upc,qty) if new sale then
<< *new* >>Sale.create()
spec:=ProductCataLog.specification(upc)
Sale.makeLineItem(spec,qty)

NB:

- OOD does do top-down functional decomposition
- also, the decomposition is more systematic
- **reason:** decomposition of the state space has already been largely done when building the class diagram.

Discussion

1. Alternative design?

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1. Alternative design?
   - create a : *LineItem* without setting the attributes
   - then get the *spec* and *price*
   - finally sets the attributes *spec* and *price*
2. Validation/justification of design
   - check each postcondition is met in the object sequence diagram

Design *finishSale*()

Recall the contract

Responsibilities: calculate *total*() and the postcondition

Name: endsale().

Responsibilities: Record that it is the end of entry of sale items, and display sale total.

Pre-conditions:

Post-conditions: *Sale.isComplete* was set to *true* (attribute modification).

Set *Sale.isComplete*

who should be responsible for setting the *isComplete* attribute of the *Sale* to true?

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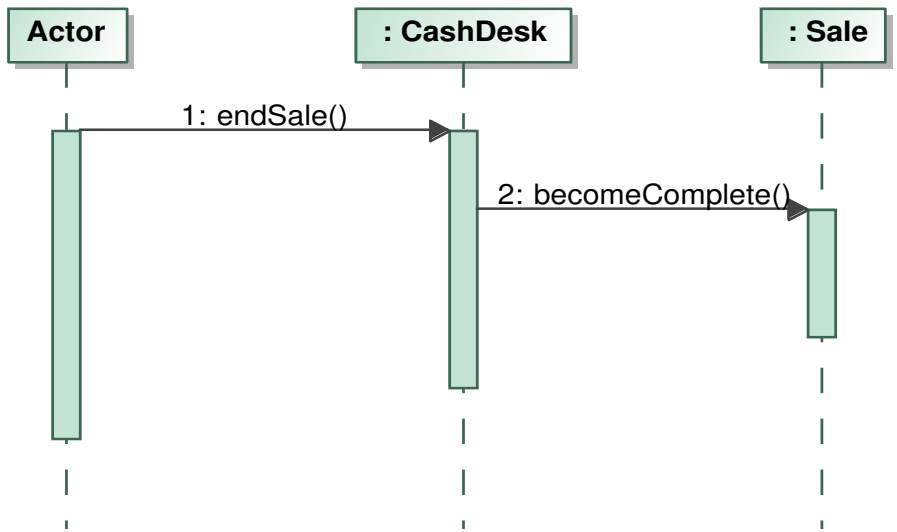

Calculating the *Sale* total

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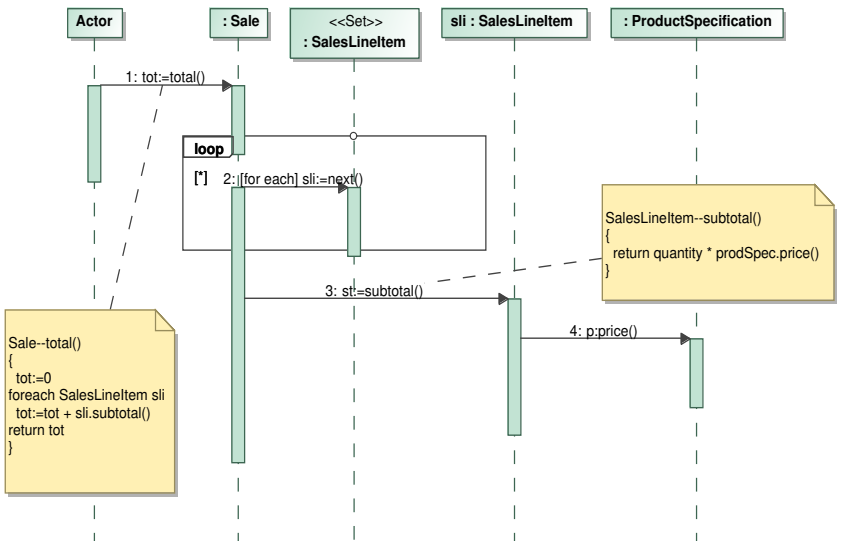

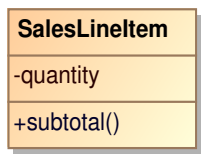

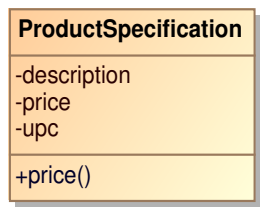

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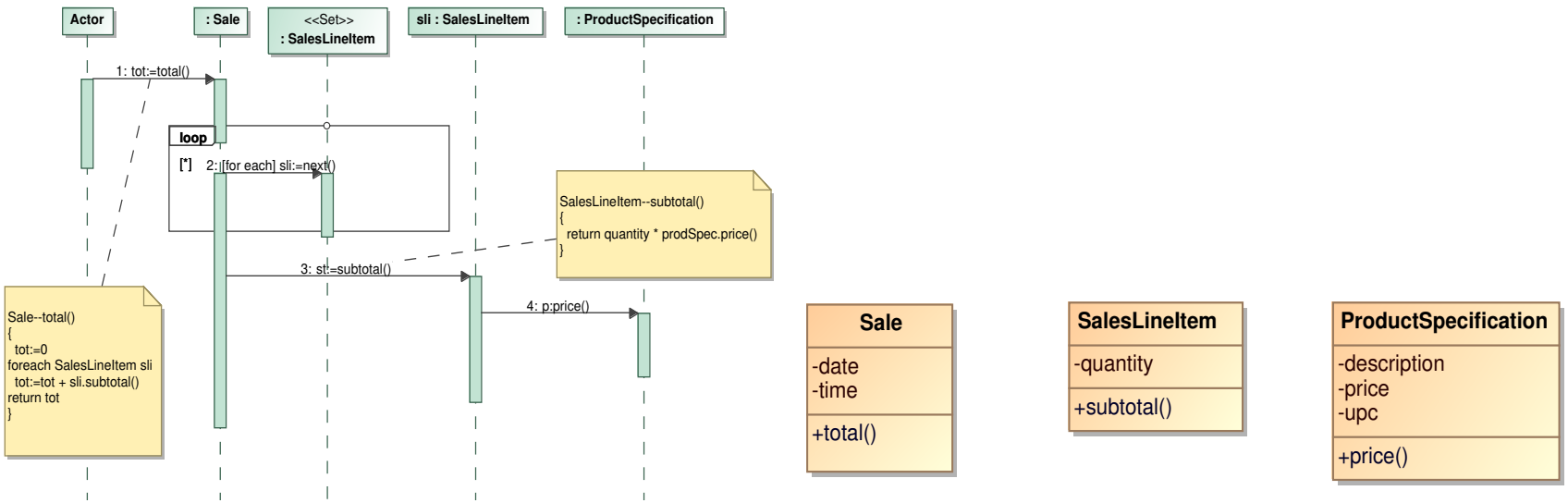

Who will send the *total* message to the *Sale*?

Design *cashPay()*

Contract

Name: makePayment(amount: Quantity).

Responsibilities: Record the payment, calculate balance and print receipt.

Post-conditions:

- A *Payment* was created (instance creation).
- *Payment.amountTendered* was set to *amount*.
- The *Payment* was associated with the *Sale*.
- The *Sale* was associated with the *Store*, to add it to the historical log of completed sales.

Design makePayment()

- We gave a design, what was it?

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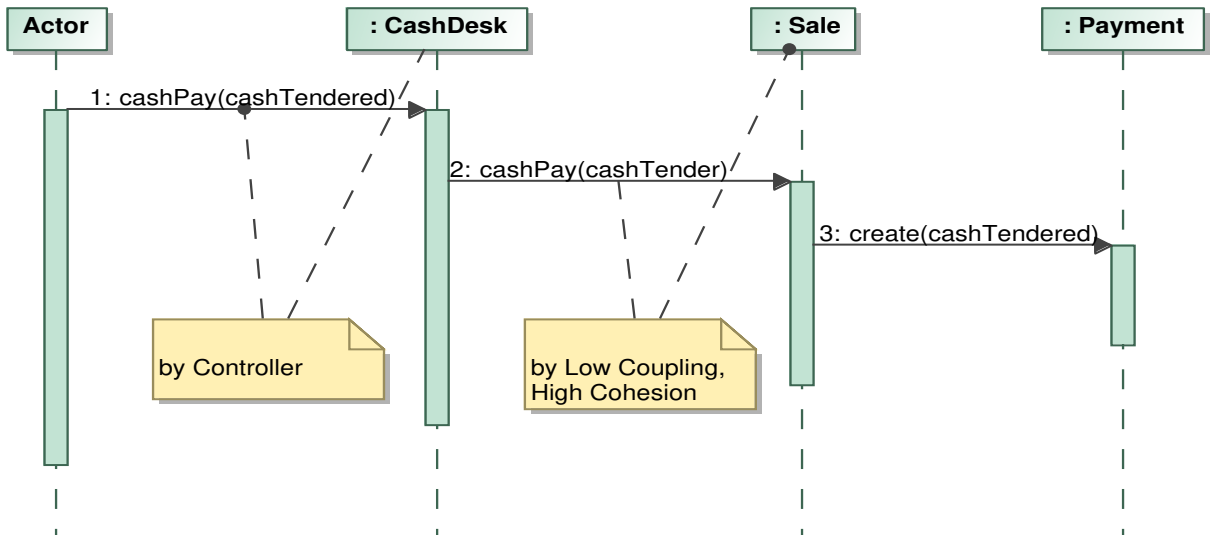
- By Creator, Low coupling and high cohesion, *Sale* is the creator of *Payment*



Logging the *Sale*

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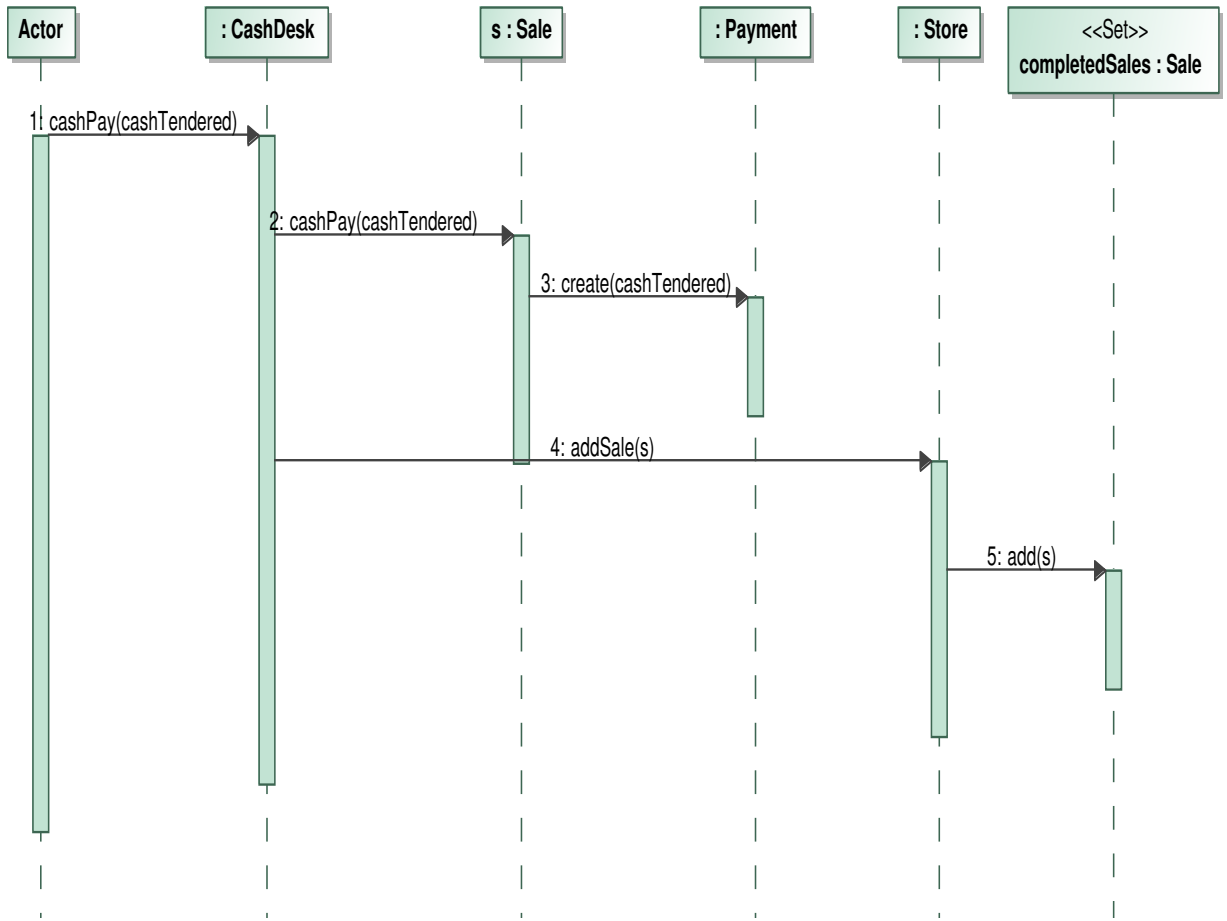

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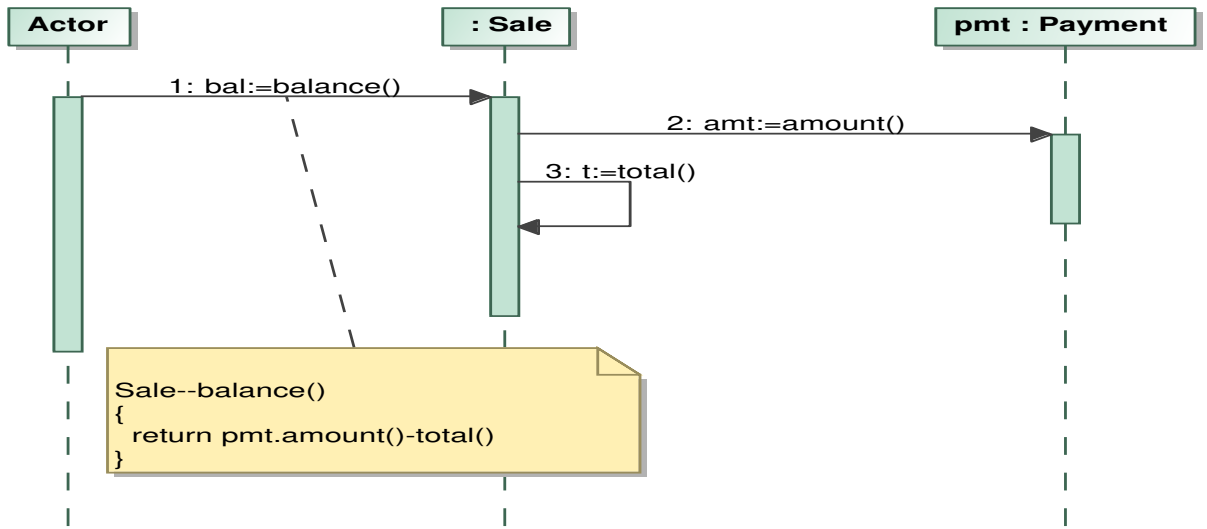


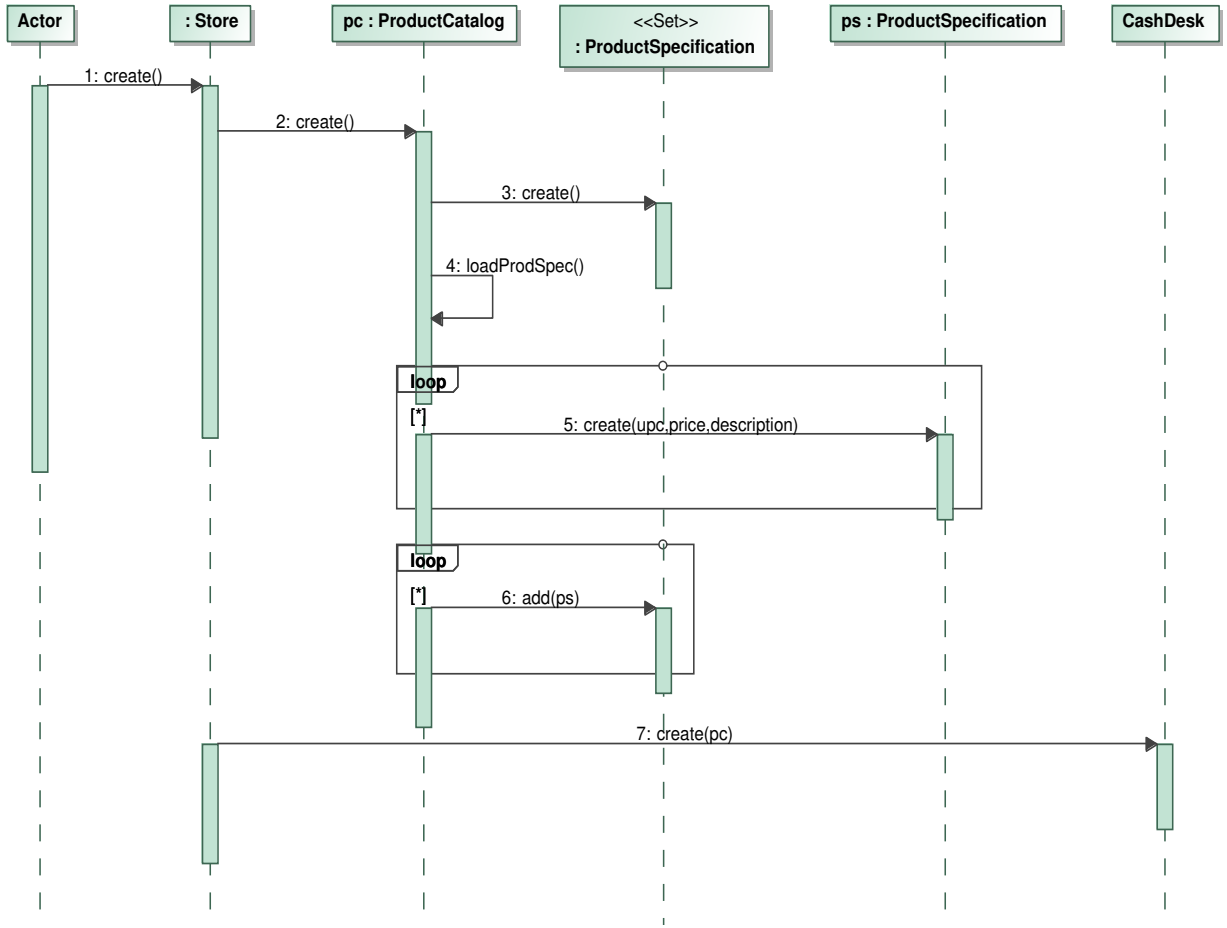
StartUp

- all components must be initialized by a *StarUp*() operation before any other operations can be used
- better to delay the design of *StartUp*() until after the other interface methods are designed

Post Conditions of *startUp*

- A Sore, CashDesk, ProductCatalog and ProductSpecification were created (instance creation).
- *ProductCatalog* was associated with *ProductSpecification* (association formed).
- *Store* was associated with *ProductCatalog* (association formed).
- *Store* was associated with *CashDesk* (association formed).
- *CashDesk* was associated with *ProductCatalog* (association formed)

Design *StartUp*()



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  - Algorithm design and design patterns
- incremental and iterative design is supported

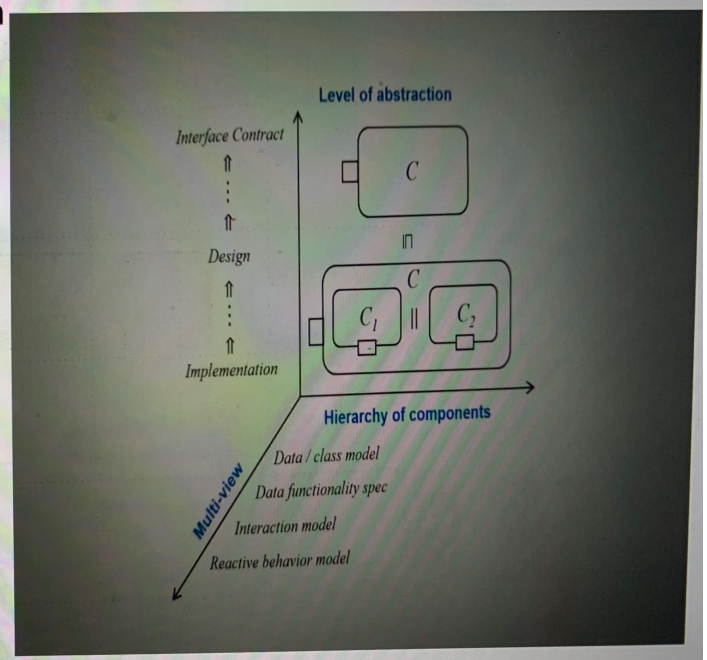
Need to practice through a whole career time

Use Case Driven Component-Based Development

Model-Driven for Predictability and Dependability

Build ***system models*** to gain confidence in requirements and designs

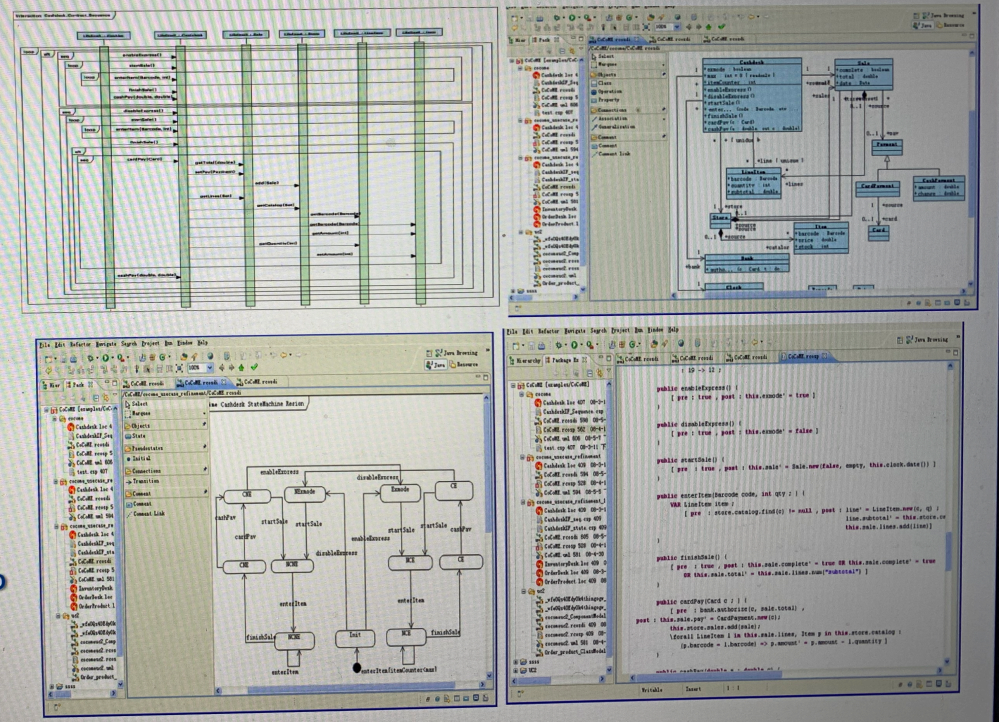
- Use **abstraction** for information hiding
  - well-focused
  - problem-oriented
- Use **decomposition** and **separation of concerns**
  - divide and conquer
  - incremental development
- use **rigor/formalization**
  - repeatable process
  - analyzable artifacts
- rCOS is fully formalizable, but allows various degree of formability in applications
- Basis for Tool Support



Use Case Driven Component-Based Development

Model–Driven Development with rCOS

- Each phase is based on the construction of ***verifiable models***
- Models are analyzed and verified
- Refined models are constructed by ***model transformations***
- ***Code*** is generated from design models
- ***Proof obligations*** are generated by model transformations
- ***rCOS modeler*** integrates UML model notation into rCOS



Model Transformations in rCOS